GRADUATION PROJECT 2026

Aqary

AI-Powered Real Estate Ecosystem & Feasibility Platform

Faculty of Computers and Data Science

Alexandria University

KEY FEATURES

- AI Chatbot
- Feasibility Engine
- Smart Recommendations
- Financial Advisory

Project Date

January 2026

Team & Supervision

SUPERVISOR

Faculty Member

Dr. Ahmed Mostafa

Department of Computer Science

Alexandria University

AA Ahmed Adel Ibrahim Ahmed

MH Mohamed Hesham Abdelwahab

AM Abdelmalek Mohamed Ahmed Mohamed

WR Wael Ahmed Mohamed Rabie El-Naqeity

AF Ahmed Farag Ahmed

KS Karim Mohamed Samy AbouShady

SS Shehab El-Din Sarwat Abdelmageed Saad

OI Omar Mohab Ibrahim

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Team Members

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Supervisor

Introduction & Overview

What is Aqary?

A web-based AI-powered ecosystem designed specifically for the Egyptian real estate market, bridging the gap between property discovery and comprehensive investment analysis.

Financial-First Search

- NLP-driven chatbot for affordability analysis
- Calculates safe installment limits
- Profiles financial health automatically

Automated Feasibility

- Estimates construction costs dynamically
- Calculates potential ROI for land investors
- Real-time material price integration

SOA Architecture

- Angular frontend with modular components
- Laravel backend for robust API services
- Python microservice for AI processing

Core Value Proposition

Democratizing access to professional-grade real estate analytics for Egyptian buyers and small-scale developers

Motivation

Economic Volatility

- Real estate as inflation hedge in Egypt
- Daily fluctuating material costs (steel, cement)
- Need for dynamic cost calculation tools

User Experience Pain

- Poor UX on existing platforms
- Lack of transparency in pricing
- Broker dependency and high fees

Technology Gap

- Existing platforms lack AI capabilities
- No advanced decision-support systems
- Missing automated feasibility tools

Engineering Solutions

- Machine learning for smart investments
- Automated feasibility modeling
- Data-driven decision support needed

Problem Statement

Fragmentation & Asymmetry

- Multiple platforms with inconsistent data
- Outdated availability and pricing info
- High broker commissions, low trust

Feasibility Blind Spot

- No tool for construction cost calculation
- Unclear ROI for land development
- Manual analysis or expensive consultants

Lack of Personalization

- Rigid search filters (Location, Price)
- Generic recommendations without learning
- Doesn't adapt to user preferences

Affordability Challenges

- Buyers struggle with affordability math
- Unclear safe installment limits
- Lack of financial transparency

Project Objectives

AI Financial Assistant

- NLP chatbot for affordability analysis
- Safe installment limits calculation

Unified Marketplace

- Residential units and lands
- C2C and B2C transaction models

Feasibility Module

- Construction cost estimation
- ROI calculation for investors

Hybrid AI Recommendations

- Content-Based Filtering for new users
- Collaborative Filtering for behavior analysis
- Personalized property suggestions

Enterprise-Grade Performance

- Scalable Angular frontend architecture
- Robust Laravel backend with Python microservices
- High traffic handling capability

Market Context & Competitive Analysis

Egyptian Real Estate Market

- Primary investment hedge against inflation
- High demand for residential and land properties
- Need for data-driven decision support tools

Aqarmap

- ✓ Massive database
- ✓ C2C model supported
- ✗ Inconsistent quality
- ✗ Lacks AI personalization

Nawy

- ✓ High-quality listings
- ✓ Excellent financing tools
- ✗ No C2C model
- ✗ Luxury market only

Bayut Egypt

- ✓ Advanced UI/UX
- ✓ Detailed filters
- ✗ Weak in land sector
- ✗ No cost estimation

Unique features that differentiate Aqary from all major Egyptian competitors

- Financial-First Search
- Feasibility Tool
- Open Marketplace
- Hybrid AI Recommendations

Literature Review: PropTech Evolution

Real Estate Evolution Journey

1.0

Static Listings Era

- Basic online classifieds
- Craigslist-style platforms
- Manual property searches

2.0

Digital Platform Era

- Mobile-first platforms
- Basic search filters
- Digital property management

3.0

AI-Powered Era

- Predictive analytics
- AI-driven valuation
- Blockchain integration

What is PropTech?

Application of information technology and platform economics to the real estate markets

Aqary Positioning

Real Estate 3.0 platform that processes data to generate financial insights and investment analysis

Literature Review: Recommendation Systems

Content-Based

- Recommends similar properties
- Based on metadata matching
- Solves cold start problem
- Limited novelty

Collaborative

- Relies on user behavior patterns
- Discovers hidden correlations
- Reveals hidden interests
- Cold start for new users

Hybrid (Aqary)

- Combines CBF and CF
- Content-Based for new users
- Collaborative for behavior
- Ensures high accuracy

Feasibility Modeling

- Dynamic material costs
- Regional labor rates
- Regulatory construction constraints

System Overview

AI-Powered Platform Beyond Traditional Listings

Bridging property discovery with comprehensive investment analysis

Conventional+

- Extends beyond listing services
- Provides analytical decision support

ROI Estimation

- Construction cost calculation
- Before land acquisition

Discovery + Analysis

- Property discovery integration
- Financial analysis capabilities

Target Users

- Individual buyers
- Small-scale developers & investors

Transparency & Intelligence

Empowers users with data-driven insights for informed investment decisions

System Architecture (SOA)

Service-Oriented Architecture

Decoupled design enabling independent scaling and robust performance

Client Layer

Angular v17 SPA

- User Interface handling
- Chatbot widget integration
- Feasibility forms & dashboards

API Gateway

Laravel 10 (PHP 8.2)

- Authentication & authorization
- Property CRUD operations
- AI service bridge communication

Intelligence Layer

Python 3.11 Microservice

- NLP parsing of user messages
- ML-powered recommendations
- Financial profiling & analysis

Data Layer

Hybrid Database Solution

- PostgreSQL for structured data
- Firebase for real-time chat
- Material prices & property data

Robust Security

High Scalability

Easy Maintenance

Technology Stack

Enterprise-Grade Technology Stack

Selected for scalability, security, and robustness in handling complex real estate data

Frontend

Angular v17

- Angular Material UI components
- TypeScript for strict typing
- Component-based architecture

Backend

Laravel 10 (PHP 8.2)

- MVC architecture pattern
- Eloquent ORM for data management
- Built-in security & authentication

AI Service

Python 3.11

- Flask API framework
- TensorFlow & Scikit-Learn
- Machine learning models

Database

Hybrid Solution

- PostgreSQL 15 for structured data
- Firebase v9 for real-time chat
- Optimized for complex queries

High Scalability

Robust Security

Proven Stability

Requirements Analysis: User Personas & Data

Data Gathering

Mixed-Method Approach

- Targeted surveys of potential buyers
- Small investors & land owners
- Social media community outreach

User Personas

Target Audience

- Individual property buyers
- Small-scale developers
- Land investors & owners

Key Survey Insights

78%

Affordability Challenge

Struggle to calculate safe installment limits

60%

Filter Inefficiency

Find standard filters ineffective

82%

C2C Preference

Prefer direct owner interaction

65%

Cost Uncertainty

Cannot estimate construction costs

Requirements Analysis: Functional Requirements

Key Functional Modules

Comprehensive system capabilities addressing Egyptian real estate market needs

Authentication & Profile

Secure User Management

- KYC verification with ID documents
- Role-based access control
- Social login integration

Property Management

Comprehensive Listings

- Advanced search & filters
- Direct property listings
- Draft & publish workflow

Feasibility Engine

Cost & ROI Calculator

- Construction cost estimation
- ROI calculation for investors
- Dynamic material price updates

AI Features

Smart Personalization

- NLP chatbot for affordability
- Hybrid recommendation system
- Behavior-based suggestions

Real-Time Chat for Secure Buyer-Seller Communication

Requirements Analysis: Non-Functional Requirements

System Quality Attributes

Critical non-functional requirements ensuring platform reliability and performance

Performance

Response Time Standards

- Chatbot response: < 1.5s
- API queries: < 200ms
- Feasibility calculation: < 2s

Scalability

Growth & Capacity

- Concurrent chat connections
- Normalized schema for 1M+ listings
- Independent scaling per layer

Security

Protection Mechanisms

- Laravel Sanctum API tokens
- Bcrypt password hashing
- XSS & CSRF prevention

Availability

Uptime Guarantee

- 99.9% uptime target
- Cloud-based hosting (AWS)
- Auto-scaling & redundancy

System Design: Architecture & Flow

Client Layer
Angular SPA

API Gateway
Laravel

Intelligence
Python AI

Data Layer
PostgreSQL

Key System Flows

AI Chatbot Flow

- User Chat Widget
- Laravel API Python NLP
- Entity Extraction
- Filter & Recommend

Feasibility Flow

- Land Details Input
- Fetch Material Prices
- Calculate Costs & ROI
- Generate PDF Report

Recommendation Flow

- User Opens Home
- Send History to Python
- ML Model Prediction
- Return Top Listings

MVC Pattern

Decoupled Design

Independent Scaling

AI Components: Chatbot

NLP-Powered Financial Assistant

Conversational AI that understands user finances and recommends affordable properties

NLP Financial Profiling

Natural Language Understanding

"I earn 30k and have 500k savings"

- Parses natural language inputs
- Extracts financial entities

Entity Extraction

Financial Data Identification

- Monthly Income
- Available Cash / Down Payment
- Existing Debts

Affordability Calculation

Safe Installment Limits

40% Rule

Installments ≤ 40% of Monthly Income

- Ensures financial security
- Prevents over-commitment

Smart Filtering

Property Recommendations

- Affordable carousel display
- Matches safe payment capacity
- Filters unaffordable options

AI Components: Recommendation System

Hybrid Filtering for Personalized Property Suggestions

Adaptive recommendation engine that learns from user behavior and preferences

Hybrid Filtering

Adaptive Algorithm

1. Content-Based
   For new users
2. Collaborative
   As data accumulates

Interaction Weights

Behavior Scoring

1	3	5
View	Save	Contact

Cosine Similarity

Attribute Matching

- Location proximity
- Price range matching
- Area & amenities comparison

Continuous Learning

Adaptive Improvement

- Tracks clicks & time spent
- Learns from saves & favorites
- Improves accuracy over time

AI Components: Feasibility Engine

Automated Construction Cost & ROI Analysis

Professional-grade feasibility calculations for land investors and developers

Cost Estimation

Dynamic Pricing Model

- Steel & cement prices daily
- Regional labor rates
- Material consumption standards

ROI Calculation

Profit Margin Analysis

Profit Margin Ratio (PMR)

- Expected selling price
- Net profit projection

Regulatory Constraints

Egyptian Building Laws

60% Footprint Ratio

- Building utilization ratios
- Max floor limitations

PDF Reports

Professional Documentation

- Skeleton cost breakdown
- Finishing cost analysis
- Net profit margin details

Implementation Highlights

Development Methodology & Environment

Agile Scrum approach with enterprise-grade technology stack

Agile Scrum

Iterative Development

- 2-week sprints
- Continuous testing
- Collaborative team workflow

Tech Stack

Enterprise-Grade Tools

- Angular v17
- Laravel 10
- Python 3.11
- PostgreSQL

Key Modules

Core Functionality

- Authentication System
- Property Management
- Feasibility Engine
- AI Chatbot
- Recommendation System

Integration

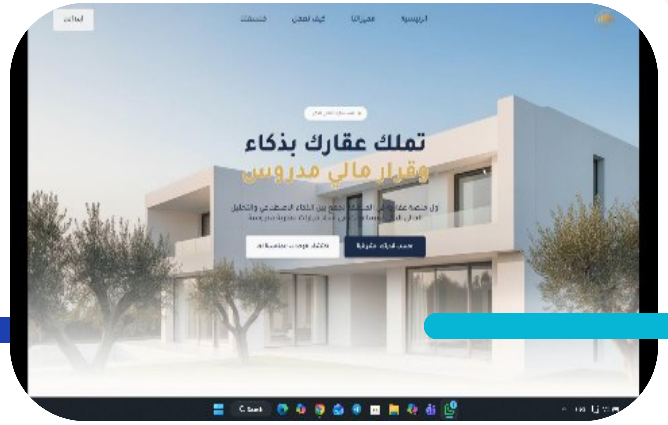
System Architecture

- REST API endpoints
- Firebase real-time chat
- Database connectivity

UI/UX Screens Showcase

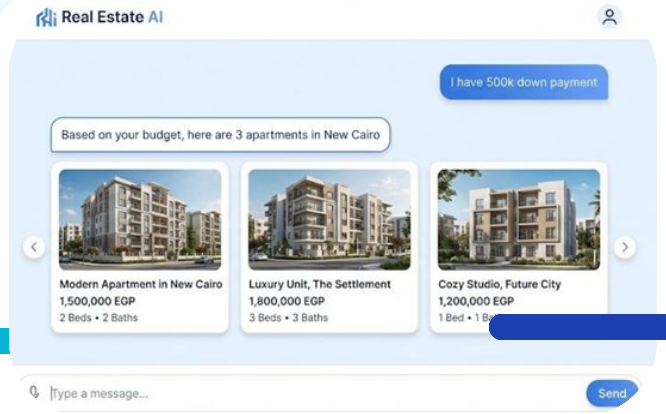
Key Interface Elements & User Flows

Intuitive design focused on accessibility and usability



Landing Page

Property services with clear calls to action for buyers and sellers

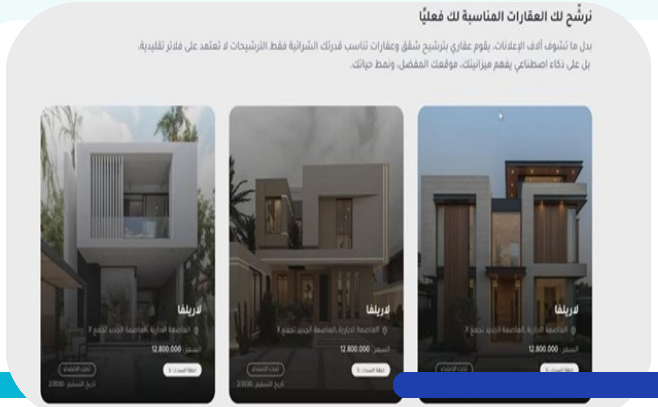


AI Chat Interface

Conversational AI for affordability analysis with property recommendations

Feasibility Dashboard

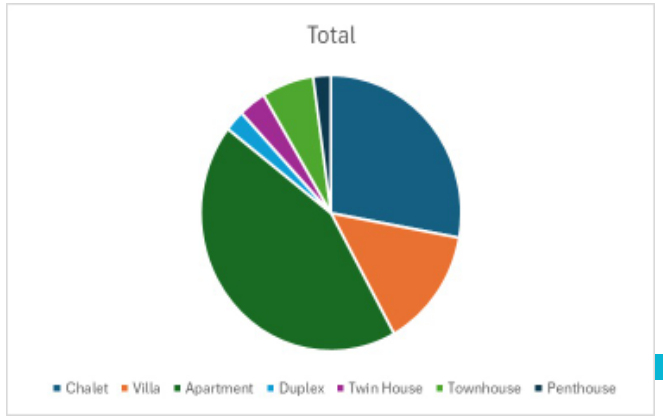
Cost breakdown visualization with investment analysis tools



Data Visualization Showcase

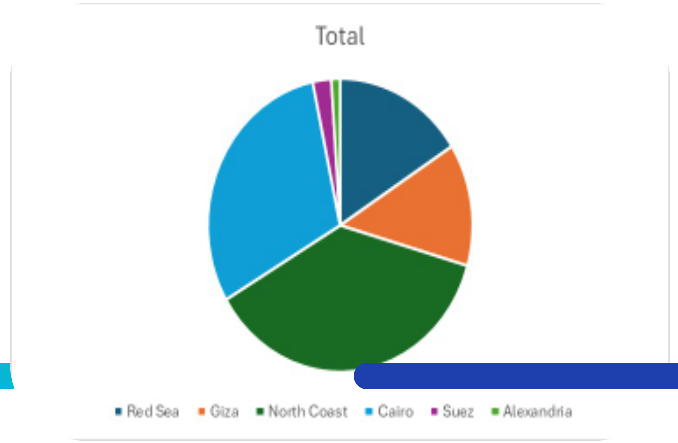
Market Insights & Property Analysis

Key visualizations revealing Egyptian real estate trends and patterns



Property Types

Apartments dominate (40%), followed by Villas and Chalets in the Egyptian market



Regional Distribution

Cairo leads with 35%, North Coast growing rapidly as secondary market

Price Analysis

Market pricing insights across regions

- Highest: North Coast & Cairo compounds
- Lowest: Suez & emerging areas
- Average: EGP 2-5M for apartments

Diverse Portfolio

Multiple property types available

Market Expansion

Growth in coastal & new cities

Price Transparency

Regional pricing patterns clear

Results & Outcomes

Key Achievements & Impact

Validation of Aqary's value proposition in the Egyptian market

Affordability Challenge

78 %

Users can now calculate safe installment limits with AI assistance

AI Trust Factor

72 %

Willing to trust AI tools for cost estimation with transparent data sources

Cost Uncertainty

65 %

Can now estimate construction costs and project profitability

Recommendation Satisfaction

High

Personalized, property-matching AI recommendations driving user engagement

Market Gap Successfully Bridged

Comprehensive Egyptian real estate solution addressing affordability, feasibility, and personalization needs

Future Work

Planned Enhancements & Roadmap

Strategic initiatives to expand Aqary's capabilities and market presence

Bank Integration

Financial Ecosystem

- Connect to CIB & NBE APIs
- Instant loan pre-approval
- Seamless mortgage application

Voice Interaction

Natural Experience

- Egyptian Arabic colloquial
- Speech-to-Text APIs
- Hands-free navigation

Construction Marketplace

Direct Procurement

- Order steel & cement directly
- Post-feasibility ordering
- Supplier network integration

Enhanced ML Models

Prediction Accuracy

- Larger training datasets
- Deep learning integration
- Real-time model updates

Mobile App Development

Native Platform Applications

- Native iOS Application
- Native Android Application
- On-the-go accessibility

Thank You

Transforming Egyptian Real Estate with Data-Driven Intelligence

Market Gap Solved

Critical need for data-driven tools in Egyptian real estate

AI Integration Success

NLP chatbot & Feasibility Engine empower investors

Scalable Architecture

SOA design suitable for enterprise growth

Paradigm Shift

From traditional "Search" to data-driven "Financial Advisory"

Enabling Egyptian users to make informed, data-backed investment decisions with confidence